

Fundamental charge: $e = 1.6 \times 10^{-19} \text{ C}$

Particles :

Properties:

- ② Total charge is Conserved

③ A charge is Quantized

→ A charge is ^{2nd} integer ^{classical} multiple of $\pm e$

20/10/21

electric charge

conductors: Materials having free electrons to move.
→ not bonded electrons

Insulators: Materials having no free electrons to move.
bonded electrons

• • → free electrons

• • • • → Valance electrons

• • • •

⊕

Current: A number of electrons passing through a conductor in 1 second

$$I = \frac{Q}{t} \text{ C/s}$$

$$1 \text{ A} \rightarrow 1 \text{ C/s}$$

number of electrons in 1 C:

$$n = \frac{Q}{e} = \frac{1 \text{ C}}{1.6 \times 10^{19}} = 6.25 \times 10^{18} \text{ C}$$

$$n = 6.253 \times 10^{18} \times$$

electric charge

$$n = \frac{q_{\text{pt}}}{e}$$
$$n = \frac{m_{\text{pt}}}{m}$$

$$\frac{q_{\text{pt}}}{e} = \frac{m_{\text{pt}}}{m}$$

- ① A group of Protons having $136 \times 10^{-19} \text{ C}$. The mass of the group is:
- $$m_{\text{pt}} = \frac{q_{\text{pt}}}{e} \times m = \frac{136 \times 10^{-19}}{1.6 \times 10^{-19}} \times 1.67 \times 10^{-27} = 141.95 \times 10^{-27} \text{ kg}$$

- ② A group of Protons having $153.64 \times 10^{-27} \text{ kg}$. The charge of the group is:
- $$\frac{q_{\text{pt}}}{e} = \frac{m_{\text{pt}}}{m} \times e = \frac{153.64 \times 10^{-27}}{1.67 \times 10^{-27}} \times 1.6 \times 10^{-19} = 147.2 \times 10^{-19} \text{ C}$$

قوة كهربائية ثابتة

electric force

electrostatic force $\longrightarrow$ between two stationary charges

Coulomb's force $\longrightarrow F_e = K_e \frac{q_1 q_2}{r_{1,2}^2}$, $K_e = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$ if a medium is air

Coulomb's force is directly proportional to the product of the two charges
" " is inversely " " square " " distance

Coulomb's force can be

① Attractive

② Repulsive

electric force

- ① $F = 100\text{ N}$. If a distance is doubled, F will be:

$$F = \frac{100}{2^2} = \underline{25\text{ N}}$$

- ② $F = 100\text{ N}$. If one of the charge is doubled but the distance halved, F will be:

$$F = 100 \times 2 \times 2^2 = \underline{800\text{ N}}$$

- ③ $F = 100\text{ N}$. If the two charge is halved but a distance change from 12 m to 36 m. F will be:

$$F = \frac{100}{2 \times 2 \times 3^2} = 100/36$$

$$\frac{36}{12} = 3$$